

Two-Period Example and Envelope Theorem

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1 Setup

Consider a two-period economy with $t = 0, 1$. Individual i receives exogenous labor income $Y_{i,0}$ at $t = 0$ and $Y_{i,1}$ at $t = 1$. There is one asset with price $P_0 > 0$ at $t = 0$ that pays a dividend $D_1 > 0$ at $t = 1$.

Individual i enters period 0 with an endowment of $N_{i,-1}$ shares and chooses holdings $N_{i,0}$ to maximize lifetime utility:

$$V_{i,0} = \max_{N_{i,0}} U(C_{i,0}) + \beta U(C_{i,1}) \quad (1)$$

subject to the budget constraints

$$C_{i,0} = Y_{i,0} + N_{i,-1}P_0 - N_{i,0}P_0, \quad (2)$$

$$C_{i,1} = Y_{i,1} + N_{i,0}D_1, \quad (3)$$

where $U(\cdot)$ is increasing and strictly concave, and $\beta < 1$ is the discount factor.

The welfare effect of a change in asset prices is measured in money-metric terms as the *equivalent variation* scaled by marginal utility:

$$\frac{dV_{i,0}}{U'(C_{i,0})}. \quad (4)$$

This expresses the welfare change in units of period-0 consumption.

2 Envelope Theorem Result

Proposition 1. *Applying the envelope theorem to the individual's problem, the welfare effect of a change in the asset price P_0 is:*

$$\frac{dV_{i,0}}{U'(C_{i,0})} = (N_{i,-1} - N_{i,0}) dP_0. \quad (5)$$

Interpretation. What matters for welfare is *net asset sales* ($N_{i,-1} - N_{i,0}$), not asset holdings. An individual who holds assets but does not trade ($N_{i,0} = N_{i,-1}$) experiences zero welfare effect from a price change — paper gains and losses are irrelevant for non-traders. Only individuals who are net sellers ($N_{i,-1} > N_{i,0}$) benefit from a price increase, and net buyers ($N_{i,0} > N_{i,-1}$) are hurt.

3 Proof via the Lagrangian

Write the Lagrangian for the individual's problem, embedding the budget constraints (2)–(3):

$$\mathcal{L} = U(Y_{i,0} + N_{i,-1}P_0 - N_{i,0}P_0) + \beta U(Y_{i,1} + N_{i,0}D_1). \quad (6)$$

Step 1: First-order condition. Differentiating with respect to the choice variable $N_{i,0}$:

$$\frac{\partial \mathcal{L}}{\partial N_{i,0}} = -P_0 U'(C_{i,0}) + \beta D_1 U'(C_{i,1}) = 0. \quad (7)$$

This is the standard Euler equation: $P_0 U'(C_{i,0}) = \beta D_1 U'(C_{i,1})$.

Step 2: Total differentiation with respect to P_0 . Totally differentiate the value function (at the optimum) with respect to P_0 . By the envelope theorem, we can ignore the indirect effect through the optimal choice $N_{i,0}^*$ (since $\partial \mathcal{L} / \partial N_{i,0} = 0$ at the optimum). This gives:

$$\frac{dV_{i,0}}{dP_0} = \frac{\partial \mathcal{L}}{\partial P_0} = (N_{i,-1} - N_{i,0}) U'(C_{i,0}). \quad (8)$$

To see this, differentiate the Lagrangian with respect to P_0 holding $N_{i,0}$ fixed:

$$\frac{\partial \mathcal{L}}{\partial P_0} = U'(C_{i,0}) \cdot (N_{i,-1} - N_{i,0}). \quad (9)$$

The term $(N_{i,-1} - N_{i,0})$ arises because P_0 appears in the period-0 budget constraint (2) multiplied by both $N_{i,-1}$ (asset endowment value) and $-N_{i,0}$ (cost of new purchases). The period-1 constraint (3) does not involve P_0 , so there is no direct effect through $C_{i,1}$.

Step 3: Money-metric welfare. Dividing both sides by $U'(C_{i,0})$:

$$\frac{dV_{i,0}}{U'(C_{i,0})} = (N_{i,-1} - N_{i,0}) dP_0, \quad (10)$$

which is the desired result (5). ■